

Adversarial Attacks on Critical Infrastructure IDS: A Water Treatment Case Study

Cybersecurity Capstone Project

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Abstract

This project investigates the robustness of Machine Learning-based Intrusion Detection Systems (IDS) in Critical Infrastructure scenarios. Replicating the methodology proposed by Barletta and del Vescovo [1] for the Automotive domain, this work applies Black-Box Adversarial Machine Learning attacks to the Water Treatment domain using the CISS 2020 dataset. We evaluate Decision Tree, Random Forest, and XGBoost models against ZOO, Boundary, HopSkipJump, and Sign-OPT attacks. Results show that while ensemble models are robust against gradient-estimation attacks, they remain vulnerable to decision-based attacks like Sign-OPT. Notably, XGBoost demonstrates superior resilience against geometric attacks compared to Random Forest. Furthermore, we analyze the trade-off between feature engineering efficiency and adversarial vulnerability.

1 Introduction

The integration of Machine Learning (ML) in Cyber-Physical Systems (CPS) has improved anomaly detection capabilities. However, ML models introduce a new attack surface: Adversarial Examples. This project aims to verify the transferability of the attack pipeline defined in [1] from the *Automotive* domain (CAN bus data) to the *Water Treatment* domain (Sensor data), analyzing the impact on widely used classifiers.

2 Methodology

2.1 Dataset: CISS 2020 (SWaT)

The dataset used is the *Critical Infrastructure Security Showdown 2020* (CISS) [2], collected from the Secure Water Treatment (SWaT) testbed. Unlike the discrete CAN bus data used in the reference thesis, this dataset contains continuous physical measurements (e.g., water levels, flow rates).

To better understand the physical dynamics of the system, Table 1 details the specific equipment (Pumps, Valves, and Sensors) involved in the Chemical Dosing stage (P2), which is the focus of our feature analysis, ultrafiltration (P3), dechlorination (P4), RO (Reverse osmosis) permeate transfer (P5).

Table 1: Complete Equipment List for SWaT Testbed (Stages P1-P6).

Stage	Tag	Description	Specification / Model
P1	FIT101	Raw Water Flow Meter	Electromagnetic DN25
	LIT101	Raw Water Tank Level	Ultrasonic, Range 0.2 to 6m
	MV101	Inlet Valve	DN 25, Electric Actuated
	P101/102	Raw Water Pumps	Duty: 2.5 m^3/h @ 20m
P2	FIT201	Feed Flow Meter	Electromagnetic DN25
	AIT201	Conductivity Analyzer	Up to 1000 $\mu S/cm$
	AIT202/203	pH & ORP Analyzer	pH: 0-14; ORP: -800 to 800mV
	MV201	Outlet Valve	DN 25, Electric Actuated
	P201/202	NaCl Dosing Pumps	50 l/h @ 10 bar
	P203-208	Chemical Dosing Pumps	HCl, NaOCl (various capacities)
P3	FIT301	UF Feed Flow Meter	Electromagnetic DN25
	LIT301	UF Feed Tank Level	Ultrasonic, Range 0.2 to 6m
	DPIT301	Differential Pressure	Range: 0-2 Bar
	MV301-304	Backwash Valves	DN 25, Electric Actuated
	P301/302	UF Feedwater Pumps	Duty: 2.5 m^3/h @ 20m
P4	FIT401	RO Feed Flow Meter	Electromagnetic DN25
	LIT401	RO Feed Tank Level	Ultrasonic, Range 0.2 to 6m
	AIT401	Hardness Monitor	Range: 0-10ppm
	AIT402	ORP Analyzer	ORP: -800mV to 800mV
	UV401	Dechlorination Unit	Removal up to 0.5ppm
	P401-404	RO Feed & Dosing Pumps	Various capacities
P5	FIT501-504	Flow Transmitters	Electromagnetic DN25
	PIT501-503	Pressure Transmitters	Range: 0-10 Bar
	AIT501-504	Water Quality Analyzers	pH, ORP, Conductivity
	P501/502	High Pressure Pumps	Duty: 2 m^3/h @ 5bar
P6	FIT601	Backwash Flow Meter	Electromagnetic DN25
	P601-603	Transfer & Cleaning Pumps	Various capacities
	LS601-603	Level Switches	Low & High Alarm

2.2 Preprocessing and Leakage Removal

A critical phase of this project was the identification and removal of **Data Leakage**. The raw dataset contained metadata columns (e.g., **Attack ID**, **Intent**) that directly revealed the class label. To simulate a realistic IDS, these columns were removed, leaving only 58 physical sensor features. Subsequently, a **Feature Engineering** phase removed constant and highly correlated features (> 0.99), reducing the dimensionality to 46 features.

2.3 Models and Attacks

We trained three models: **Decision Tree (DT)**, **Random Forest (RF)**, and **XGBoost (XGB)**. Using the *Adversarial Robustness Toolbox* (ART) [3], we executed four Black-Box evasion attacks:

- **ZOO**: Zeroth Order Optimization (Gradient estimation).

Feature	FIT101	LIT101	MV101	Label
Sample 1	0.896	0.652	1.0	0 (Normal)
Sample 2	0.000	0.647	0.5	0 (Normal)
Sample 3	0.507	0.420	0.5	1 (Attack)

Table 2: Example of refined dataset (Normalized 0-1).

- **HopSkipJump:** Boundary-based attack (Geometric).
- **Boundary Attack:** Random walk optimization along the decision boundary.
- **Sign-OPT:** Sign-based gradient estimation (Adapted for binary classification).

3 Exploratory Data Analysis (EDA)

Before training the models, a comprehensive statistical analysis was conducted to understand the underlying data distribution and the physical relationships between the sensors of the Water Treatment plant.

3.1 Feature Correlation and Physical Constraints

The Correlation Matrix, shown in Figure 1, highlights strong linear relationships (Pearson coefficient > 0.98) between specific clusters of sensors. From a physical perspective, this redundancy is expected in a hydraulic process due to mass conservation laws and control logic (e.g., if a pump starts, the downstream flow rate must increase immediately).

However, from a Machine Learning perspective, this **multicollinearity** introduces redundancy without adding information gain. **Strategic Decision:** We performed a Feature Engineering step to remove these highly correlated features. This reduces the dimensionality of the search space for adversarial optimization algorithms (like ZOO), theoretically accelerating the generation of adversarial examples, as discussed in Section 6.

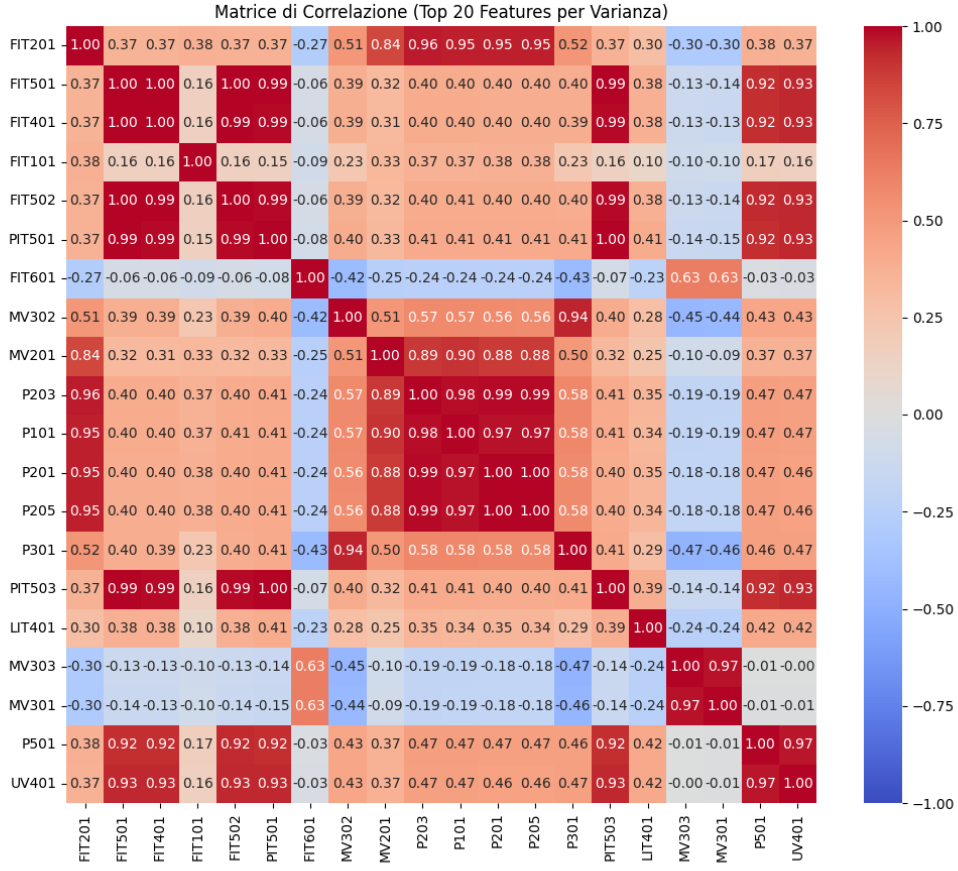


Figure 1: Correlation Matrix: Red blocks indicate redundant physical sensors (e.g., FIT501 vs UV401).

3.2 Class Distribution and Imbalance

The analysis of the target variable reveals a significant class imbalance, as shown in Figure 2. The dataset contains approximately 30,000 *Normal* instances against only 2,000 *Attack* instances (a ratio of roughly 15:1).

This imbalance reflects real-world Operational Technology (OT) environments, where cyber-attacks are rare events compared to normal operations. Consequently, relying solely on *Overall Accuracy* would be misleading (a dummy classifier predicting always "Normal" would achieve $\approx 94\%$ accuracy). Therefore, this study prioritizes **F1-Score (Weighted)** and **Recall** as the primary metrics to evaluate the IDS effectiveness.

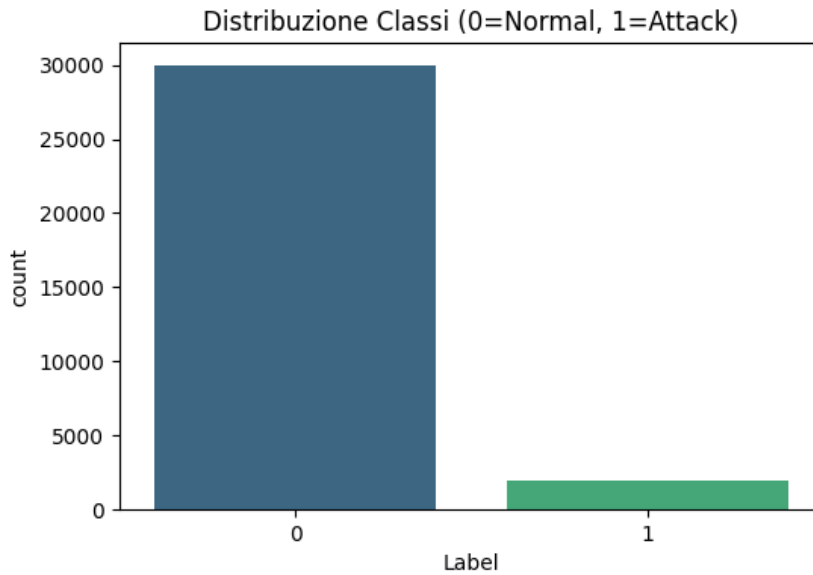


Figure 2: Class Distribution: The dataset is highly imbalanced, necessitating Weighted F1-Score evaluation.

3.3 Sensor Distribution Analysis

To further investigate the discriminative power of individual sensors, we analyzed the distribution of the top features using boxplots, as shown in Figure 3. The plots compare the range and quartiles of sensor readings during *Normal* (0) and *Attack* (1) states. It is observed that for many critical sensors (e.g., FIT201, FIT501), the distributions overlap significantly. This overlap confirms that simple threshold-based detection is insufficient for this domain, justifying the use of complex ensemble models like Random Forest and XGBoost that can capture non-linear correlations between multiple sensors.

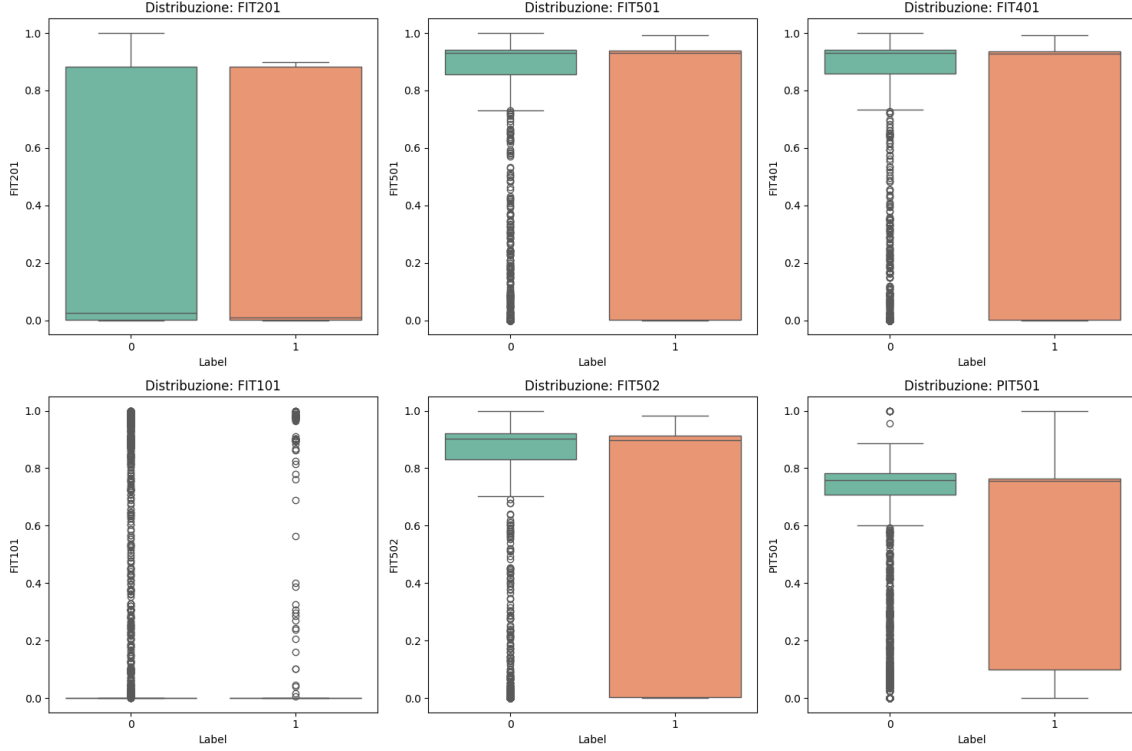


Figure 3: Boxplots of representative sensors. The overlapping distributions between classes highlight the complexity of the detection task.

4 Experimental Results

4.1 Baseline Performance

In normal conditions, ensemble models (RF, XGB) achieved superior performance compared to the single Decision Tree.

Model	F1-Score (Weighted)	Overall Accuracy
Decision Tree	0.884	88.2%
Random Forest	0.903	92.6%
XGBoost	0.907	93.4%

Table 3: Baseline performance on Test Set C.

4.2 Adversarial Impact (All Attacks)

We evaluated the models against four different attack strategies. Table 4 presents the comprehensive results.

Model	Attack	F1-Score (Adv)	Drop	Time (s)	Similarity
Decision Tree	ZOO	0.673	-23%	3.40	0.999
	Boundary	0.193	-78%	8.75	0.963
	HopSkipJump	0.193	-78%	12.27	0.979
	Sign-OPT	0.193	-78%	18.86	0.999
Random Forest	ZOO	0.841	-6%	144.08	0.997
	Boundary	0.286	-68%	133.28	0.921
	HopSkipJump	0.315	-65%	572.48	0.922
	Sign-OPT	0.011	-99%	793.67	0.916
XGBoost	ZOO	0.817	-9%	13.50	0.996
	Boundary	0.850	-6%	7.01	0.992
	HopSkipJump	0.871	-4%	8.07	0.999
	Sign-OPT	0.421	-53%	63.63	0.914

Table 4: Comprehensive attack results. While Random Forest collapses under decision-based attacks (Sign/Bound/Hop), XGBoost maintains high resilience against Boundary and HopSkipJump.

4.3 Visual Analysis: Confusion Matrices

The following figures illustrate two contrasting scenarios: the catastrophic failure of Random Forest under Sign-OPT, and the remarkable resilience of XGBoost under HopSkipJump.

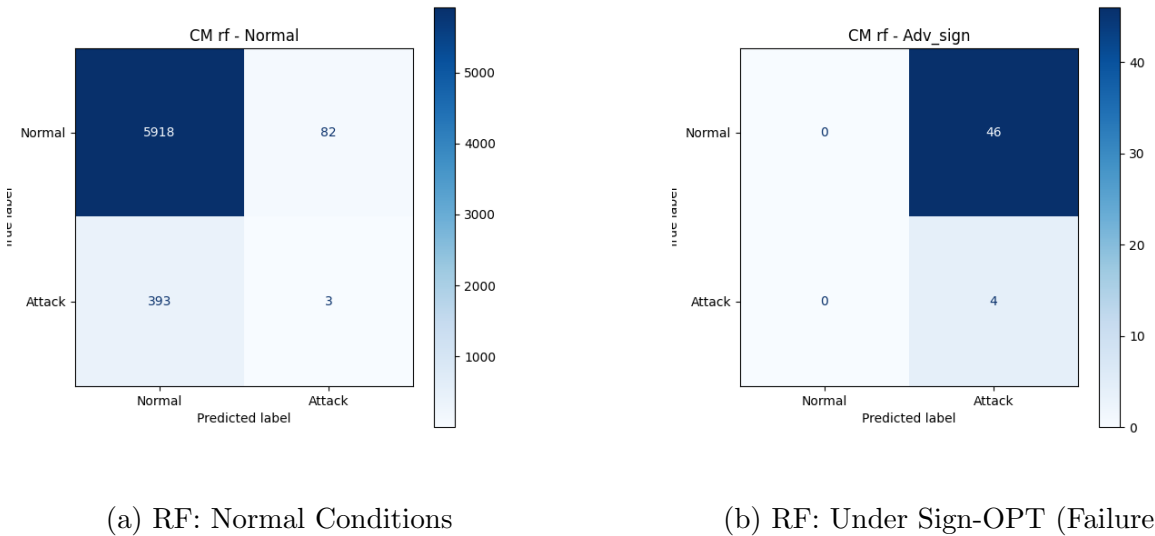
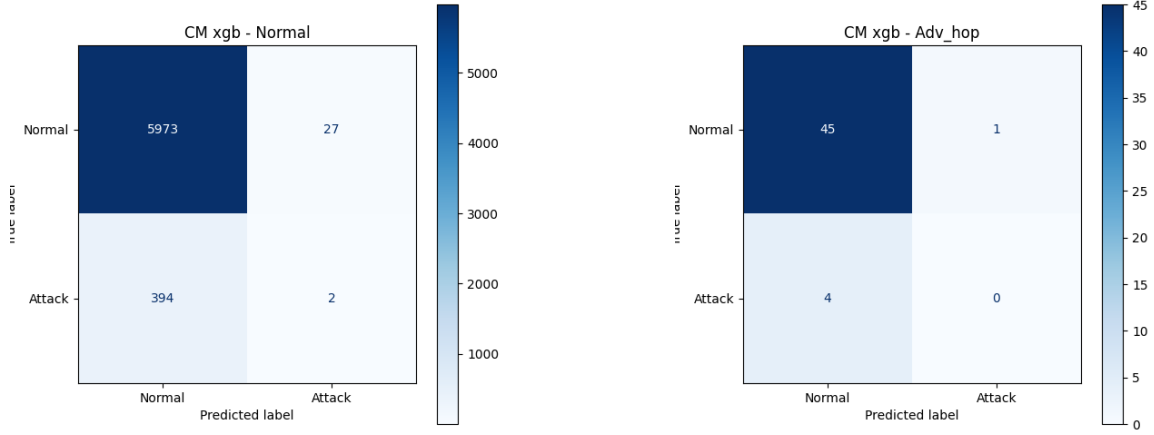


Figure 4: **Random Forest Failure:** Under Sign-OPT attack, the model fails to detect almost all attacks (False Negatives).



(a) XGB: Normal Conditions

(b) XGB: Under HopSkipJump (Robust)

Figure 5: **XGBoost Resilience:** Unlike RF, XGBoost maintains high detection rates even under HopSkipJump attack.

5 Discussion: Efficiency vs. Security

A core contribution of this study is the evaluation of how **Feature Engineering** impacts both the operational efficiency of the IDS and its adversarial vulnerability. We compared the *Processed* dataset (58 features) with the *Refined* dataset (46 features).

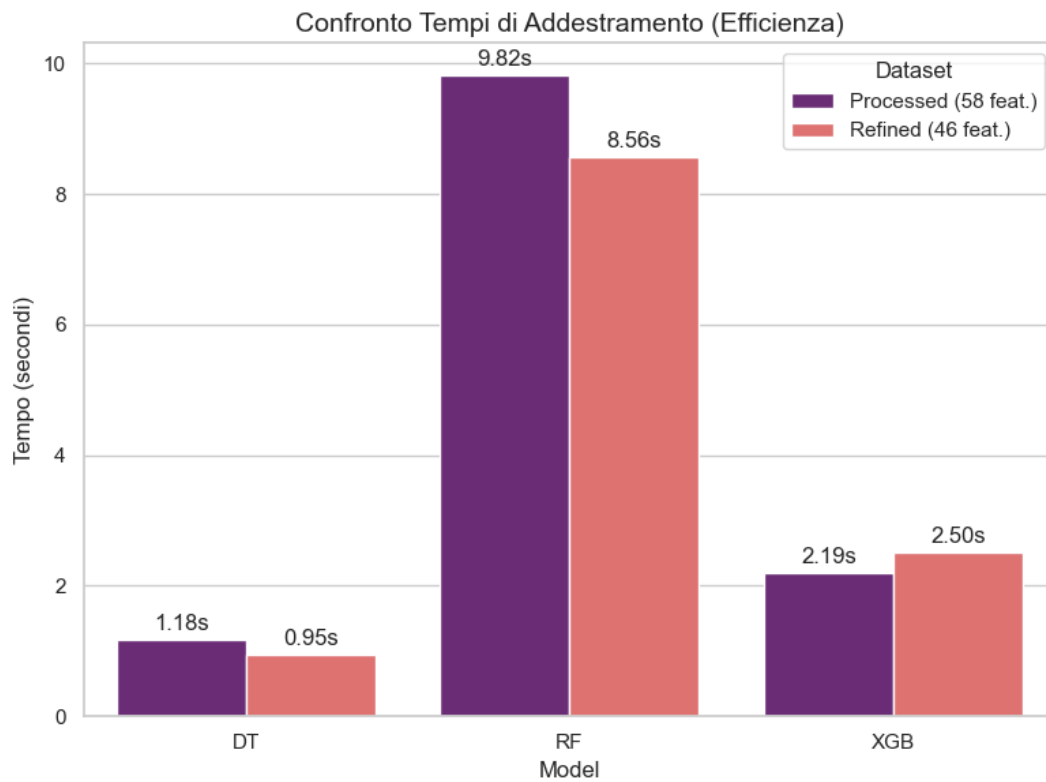
5.1 Impact on Performance and Training Efficiency

As illustrated in Figure 6a, the removal of 12 redundant and highly correlated features did not degrade the models' predictive capabilities. In fact, the Decision Tree (DT) showed a slight improvement in F1-Score, likely due to the reduction of noise in the splitting process.

Furthermore, Figure 6b demonstrates a clear gain in computational efficiency. The training time for the Random Forest model decreased by approximately 13% (from 9.82s to 8.56s). In a real-world deployment on edge devices or embedded systems within a water plant, this reduction translates to lower power consumption and faster model updates.



(a) F1-Score Stability



(b) Training Time Efficiency

Figure 6: Comparison between Processed and Refined datasets. Feature selection maintains performance while improving training speed.

5.2 The Security Paradox

Despite the gains in efficiency, the reduction of the feature space introduced a significant security trade-off. As shown in Figure 7, the time required for the ZOO attack to generate successful adversarial examples for the Decision Tree reduced by 28% (from 4.72s to 3.40s).

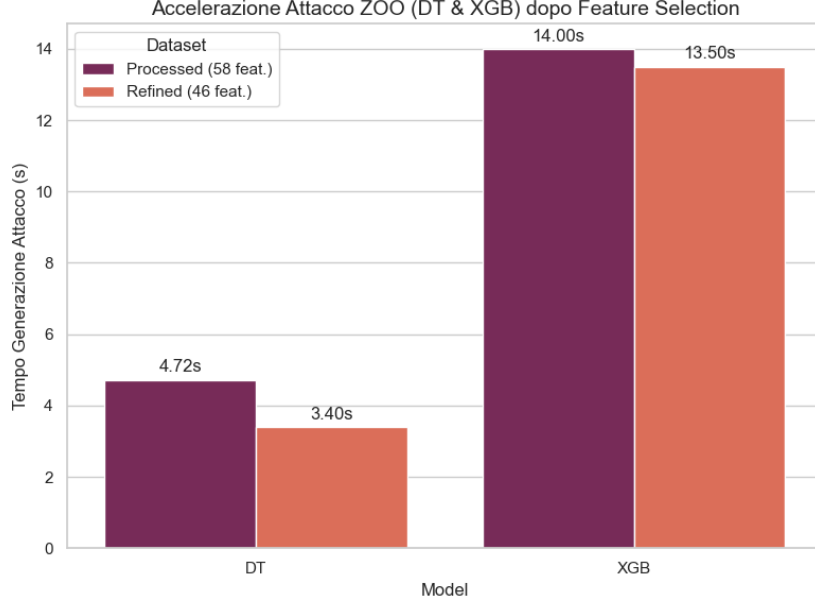


Figure 7: Acceleration of ZOO attack generation after feature selection.

This result reveals a **Security Paradox**: by optimizing the dataset for the defender (removing redundancy), we simultaneously optimized the search space for the attacker. With fewer dimensions to perturb, adversarial optimization algorithms converge significantly faster, reducing the "window of opportunity" for the defender to detect the ongoing attack generation process.

5.3 Mitigation Strategy: Adversarial Obfuscation

Based on the findings in Section 5.2 regarding the "Security Paradox" (where reducing features accelerated attacks), we proposed and tested a counter-intuitive mitigation strategy named **Adversarial Obfuscation** using "Trap Features".

5.3.1 Methodology

We hypothesized that if reducing the feature space accelerates the attack, artificially expanding it with irrelevant features (noise) should slow it down without affecting detection accuracy. We generated a *Trapped* dataset by injecting 20 columns of random Gaussian noise into the *Refined* dataset (increasing dimensionality from 46 to 66 features). We then re-trained the models and executed the ZOO attack.

5.3.2 Results

The experiment confirmed the hypothesis. As shown in Figure 8, the models successfully ignored the noise (F1-Score remained stable), while the ZOO attack algorithm was forced to waste computational resources optimizing irrelevant variables.

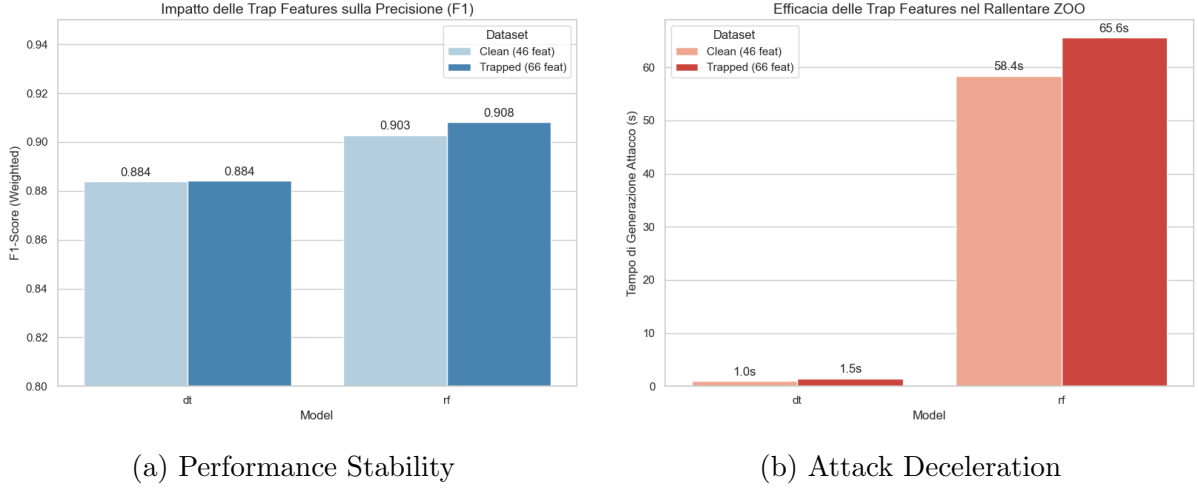


Figure 8: Effectiveness of Adversarial Obfuscation. Adding noise features slows down the attacker (Red) without degrading the defender’s accuracy (Blue).

Table 5 quantifies the impact. The strategy proved particularly effective for the Decision Tree, increasing the attack generation cost by over **50%**.

Model	Dataset	F1-Score	Attack Time (s)	Delay Induced
Decision Tree	Clean (46)	0.884	0.99	-
	Trapped (66)	0.884	1.49	+50.5%
Random Forest	Clean (46)	0.903	58.42	-
	Trapped (66)	0.908	65.63	+12.3%

Table 5: Impact of Trap Features. The attack time increased significantly with zero cost to detection performance.

This demonstrates that **Security by Obfuscation**, often dismissed in traditional cybersecurity, can be a valid defense layer in Adversarial Machine Learning against Black-Box optimization attacks.

5.4 Quantitative Analysis: The Vulnerability Curve

To rigorously validate the relationship between dimensionality and security, we conducted an advanced stress test using Recursive Feature Elimination (RFE) with increased attack iterations ($N = 100$). We measured both the Attack Generation Time and the Attack Success Rate (ASR).

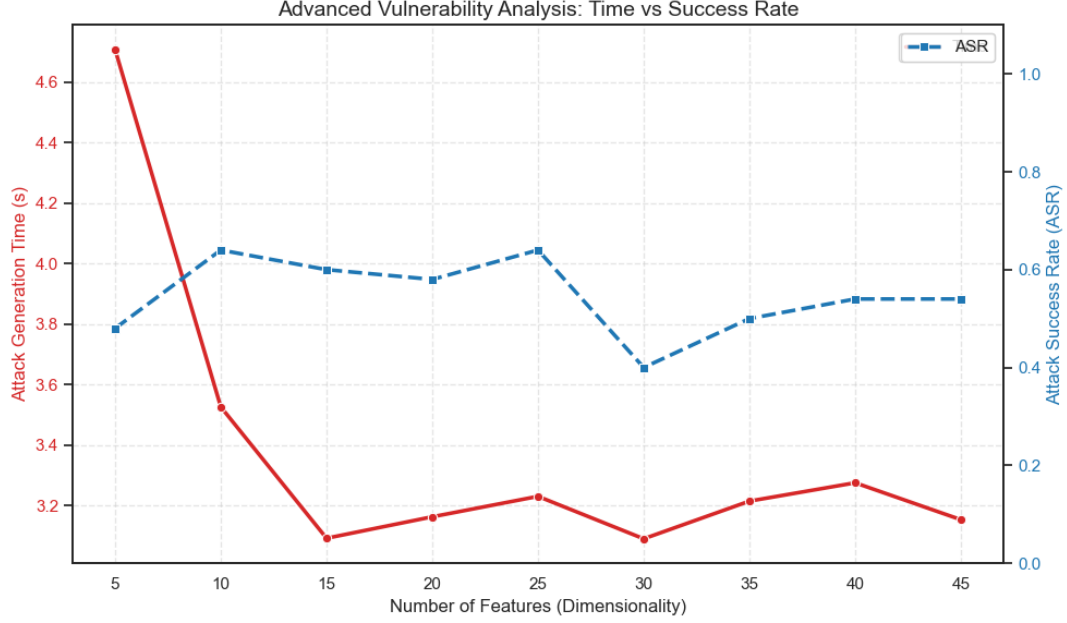


Figure 9: Advanced Vulnerability Analysis. The Red line (Time) shows a sharp drop after 5 features, confirming that increasing dimensionality initially simplifies the attack. The Blue line (ASR) peaks in the 10-25 feature range, identifying a "Sweet Spot" for the attacker.

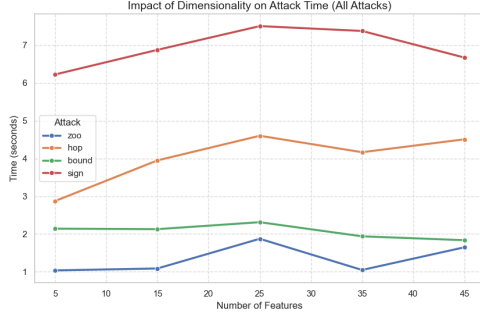
The results (Figure 9) reveal a non-linear dynamic:

- **Low Dimensionality (5 features):** The system is robust. The attack is slow (4.71s) and has a low success rate (48%), likely due to the lack of "degrees of freedom" for the perturbation.
- **The Attacker's Sweet Spot (10-25 features):** As features increase, the attack time drops significantly (≈ 3.1 s) while the Success Rate peaks (64%). The added dimensions provide shortcuts for the optimization algorithm.
- **High Dimensionality:** Beyond 30 features, the ASR stabilizes or drops, suggesting that the complexity starts to hinder the attacker again.

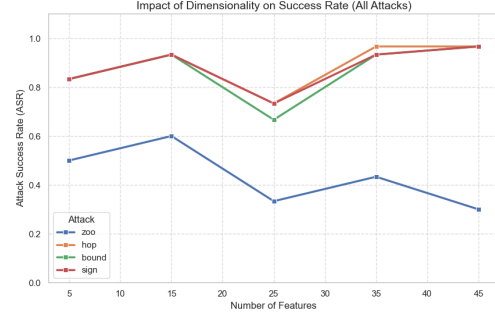
This confirms the **Security Paradox**: simplifying a model (e.g., from 58 to 46 features) might inadvertently push it into the "Sweet Spot" where attacks are most efficient. Conversely, the **Adversarial Obfuscation** strategy (adding noise) pushes the system towards high dimensionality, recovering security.

5.5 Comparative Vulnerability Analysis

We extended the dimensionality analysis to all four attacks (ZOO, HopSkipJump, Boundary, Sign-OPT) to verify if the "Security Paradox" is universal.



(a) Attack Generation Time



(b) Attack Success Rate (ASR)

Figure 10: Comparative analysis. HopSkipJump (Orange) shows a linear time increase with dimensionality. Sign-OPT (Red) is the slowest but highly effective. ZOO (Blue) is fast but unreliable (low ASR).

The results (Figure 10) reveal distinct behaviors:

- **HopSkipJump (Geometric):** Shows a clear linear increase in time (from 2.8s to 4.5s) as features grow. This confirms that geometric attacks are the most sensitive to the "Adversarial Obfuscation" defense.
- **Sign-OPT (Decision-based):** It is the most computationally expensive attack (≈ 7 s) but maintains a very high Success Rate ($> 90\%$), making it a persistent threat despite defenses.
- **ZOO (Gradient-based):** While fast (≈ 1.5 s), its Success Rate drops significantly as dimensionality increases (down to 30% at 45 features), suggesting it struggles in high-dimensional spaces.

5.6 Transferability Analysis (Gray-Box Scenario)

Finally, we evaluated the threat in a Gray-Box scenario to assess if an attacker could bypass the IDS without knowing the specific target model. We trained a surrogate model (**Decision Tree**) to generate adversarial examples using HopSkipJump and launched them against the target ensemble models (**Random Forest** and **XGBoost**).

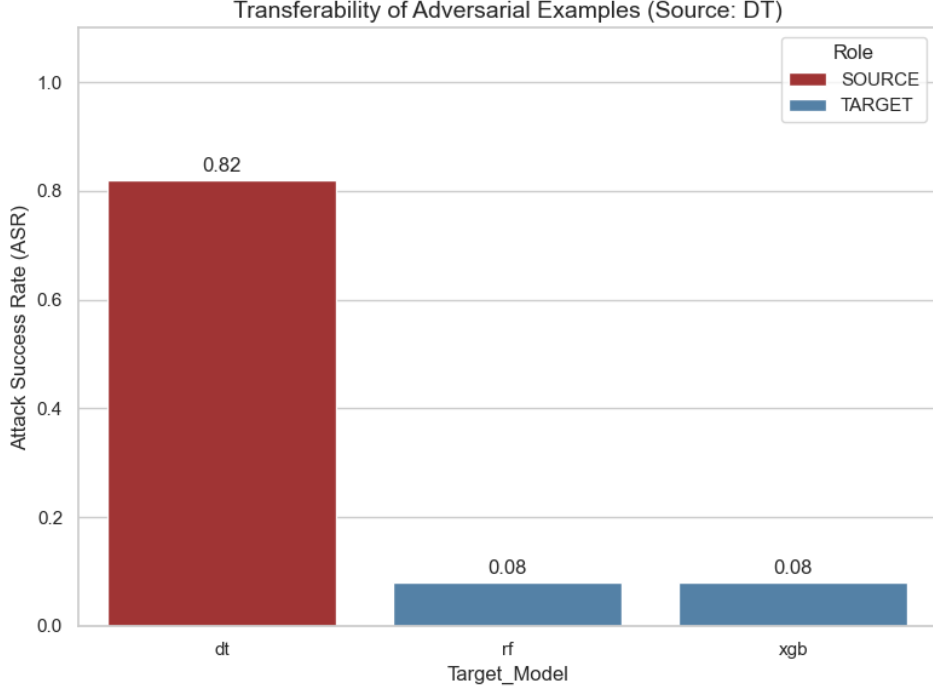


Figure 11: Transferability of adversarial examples generated on a Decision Tree. The attack succeeds on the source (Red) but fails completely on the targets (Blue).

The results (Figure 11) indicate **low transferability**:

- **Source (DT):** The attack is highly effective (ASR = 0.82).
- **Targets (RF/XGB):** The ASR drops drastically to **0.08**. This value is comparable to the natural error rate of the models, implying the attack had negligible effect.

Conclusion: Ensemble models in the Water Treatment domain exhibit intrinsic robustness against transfer attacks generated by simpler surrogate models. An attacker would need a surrogate model with an architecture much closer to the target to succeed.

6 Active Defense: Adversarial Training

To mitigate the vulnerabilities identified in Section 5, particularly the catastrophic failure of Random Forest against Sign-OPT, we implemented an **Adversarial Training** strategy. We augmented the training set with the adversarial examples generated during the attack phase and retrained the models.

6.1 Results

The results demonstrate the effectiveness of this active defense mechanism. As shown in Figure 12, the models recovered from their vulnerabilities.

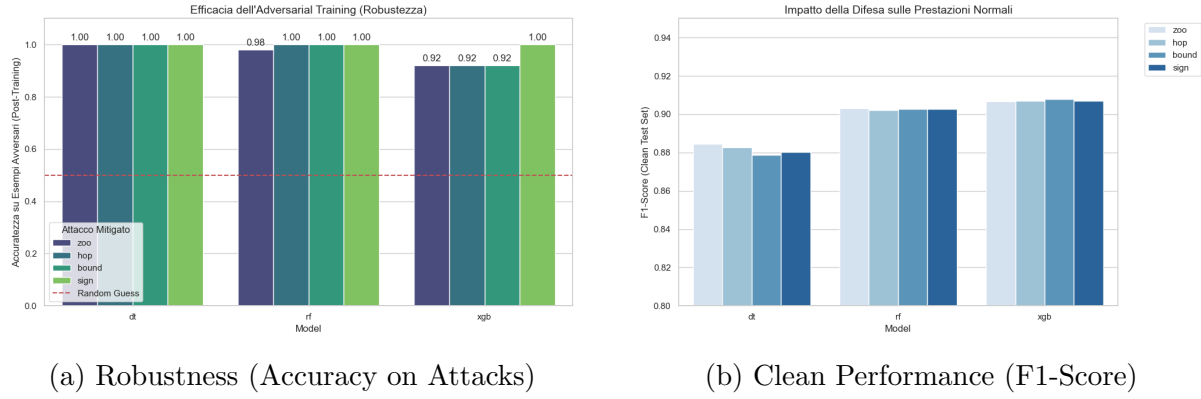


Figure 12: Impact of Adversarial Training. (a) Models achieve near-perfect detection of attacks that previously bypassed them. (b) Performance on normal data remains stable, indicating no "catastrophic forgetting".

- **Random Forest Recovery:** Before defense, RF failed completely against Sign-OPT ($F1 \approx 0.01$). After retraining, it achieves **100% accuracy** on the same adversarial examples.
- **Stability:** The performance on clean data (Normal Test Set) remained stable ($F1 > 0.90$ for RF and XGB), proving that the models learned to generalize the attack patterns without compromising normal operations.

7 Explainable AI (XAI) Analysis

To move beyond aggregate metrics and understand the mechanics of the evasion attacks, we employed **SHAP (SHapley Additive exPlanations)** to interpret the adversarial examples generated by HopSkipJump against the XGBoost model.

7.1 Perturbation Analysis

Figure 13 reveals the physical footprint of the attack. The bar chart shows the top 10 features with the highest mean absolute perturbation. The attack algorithm heavily targeted **AIT202** (Chemical Analyzer) and **FIT601** (Flow Transmitter). Interestingly, the attack also manipulated **P101** (Pump 101). In a Cyber-Physical System, modifying a pump's state is a significant operational change, suggesting that while the attack is mathematically valid against the ML model, it might create physical anomalies visible to human operators.

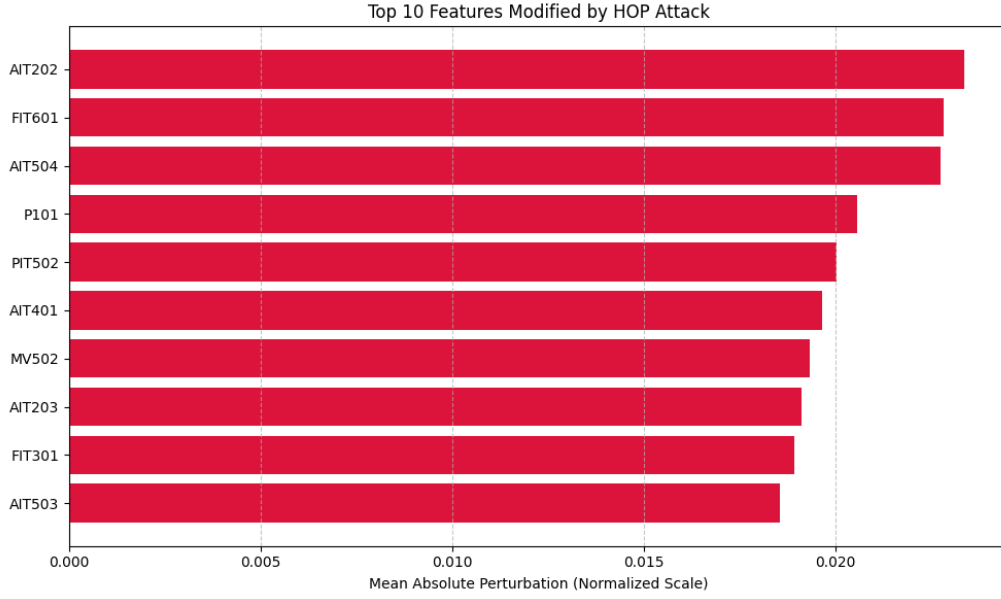


Figure 13: Top 10 sensors modified by the HopSkipJump attack. The algorithm focused its effort on chemical analyzers (AIT) and flow sensors (FIT).

7.2 Forensic Analysis of Evasion

The SHAP Waterfall plot (Figure 14) provides a forensic breakdown of a single successful evasion instance.

- **Red bars:** Features pushing the prediction towards "Attack" (e.g., FIT503, FIT504).
- **Blue bars:** Features pushing the prediction towards "Normal" (e.g., FIT101, FIT301).

The plot demonstrates that the sensor **FIT101** played a crucial role in the evasion. By manipulating or maintaining this specific flow value at 0.984, the attacker generated a strong "Normal" signal (Blue bar of -0.45) that suppressed the anomalies detected in other sensors, forcing the final classification to "Normal".

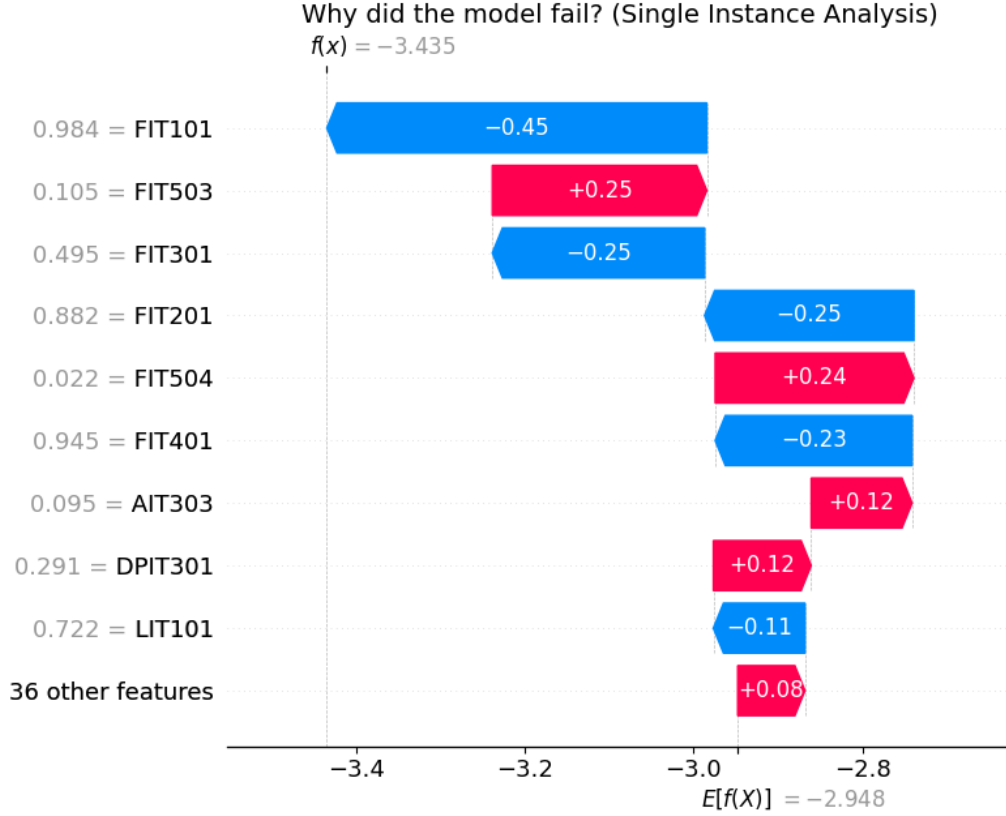


Figure 14: SHAP Waterfall plot of a successful evasion. The strong negative contribution of FIT101 (Blue) counteracts the attack indicators (Red), deceiving the model.

7.3 Global Feature Importance

Finally, Figure 15 summarizes the global feature importance. The model relies heavily on **AIT201** and **LIT401**. Comparing this with the perturbation plot (where AIT202 was the most modified), we observe that the attacker exploited secondary features to bypass the decision boundary, likely because the primary features were harder to perturb without violating physical constraints.

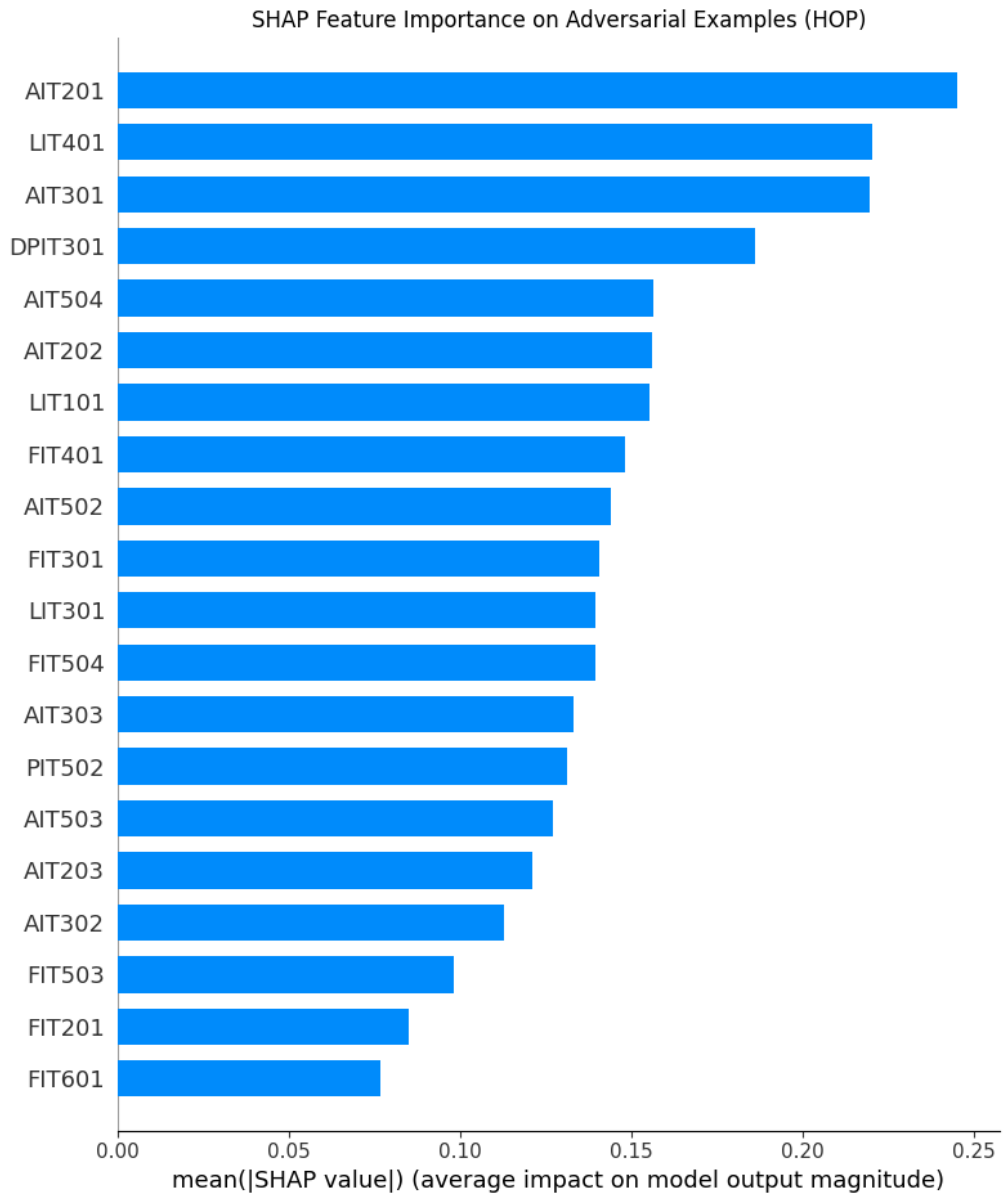


Figure 15: SHAP Summary Plot. The model primarily looks at AIT201 and LIT401 to make decisions.

8 Conclusion

This project successfully transferred the adversarial attack methodology from the Automotive to the Water Treatment domain.

- **Sign-OPT** emerged as the most dangerous attack, capable of bypassing Random Forest defenses ($F1 \approx 0$), unlike in the reference thesis where ZOO was the primary focus.
- **XGBoost** proved to be the most balanced model. It showed exceptional resilience against geometric attacks (Boundary/HopSkipJump) where Random Forest failed, although it remains vulnerable to Sign-OPT.
- **Security Trade-off:** Feature engineering improves efficiency but can reduce the computational cost for the attacker.

References

- [1] Vita Santa Barletta and Samuele del Vescovo. “Adversarial Attacks on IDS and Multidomain Impact Analysis for Threat Intelligence in Military Automotive Scenarios”. Reference Thesis. Software Engineering Research Lab, 2024.
- [2] iTrust. *CISS 2020 Dataset: Critical Infrastructure Security Showdown*. https://itrust.sutd.edu.sg/itrust-labs/datasets/dataset_info/. Secure Water Treatment (SWaT) Testbed. 2020.
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